



Focus on innovation.

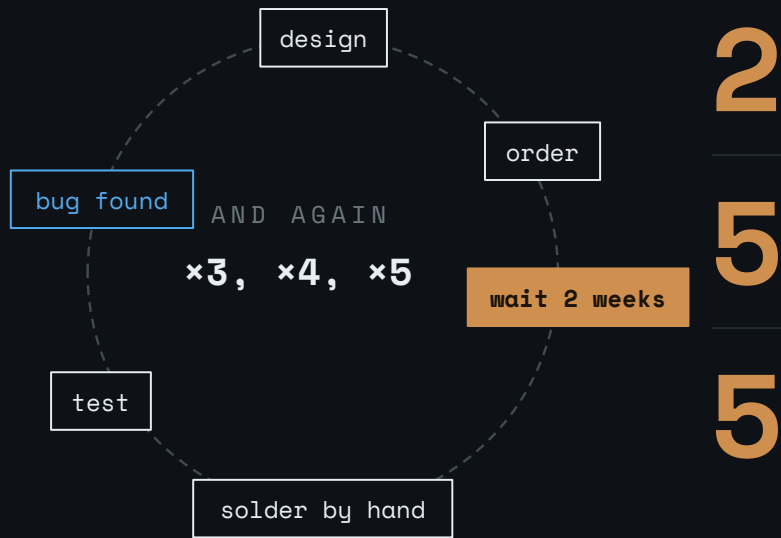
We take care of the rest.

`proto`, our AI engineer, plans every step.

`baker` makes the board. `mixer` tests it.

`protopus-kappa.vercel.app`

A prototype is a loop, and every lap costs weeks



weeks waiting for a board to come back from a fab

instruments to buy before you can measure it

vendor apps and driver stacks to keep alive

Nobody is short of ideas. **They are short of tries.**

AI already builds software. In hardware, it stops at the file.

SOFTWARE

\$100M ARR

Lovable, 8 months after launch,
selling to people who can't code.

HARDWARE FILES

~\$100M

raised by AI that draws boards: Flux,
Quilter, Diode. Claude now drives
Fusion and Blender.

THE BENCH

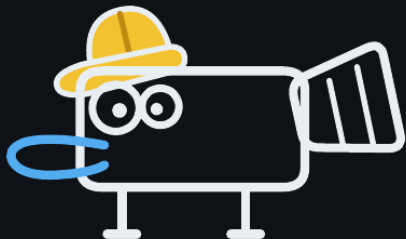
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AI tools that can make the board,
power it and see if it works.



Smart machines give the AI
access to the real world.

One AI engineer, and the two smart machines it runs



proto

Plans the build, runs the machines, explains the results.



baker

Makes the board: mill, mask, stencil, paste, reflow.



mixer

Tests it: eight instruments on one cable.

FOR PROTOTYPES · the first 1-10 boards of a design. Volume goes to a fab.

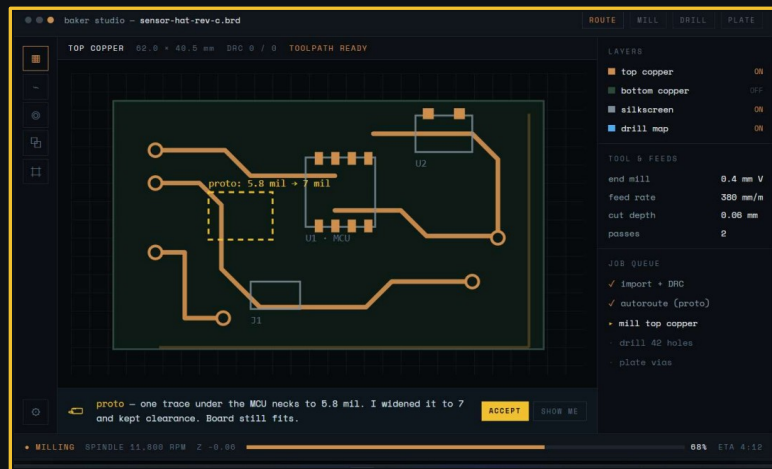
DEMO 1

Website · protopus-kappa.vercel.app

DEMO 2

Video · Baker prototype running

An AI engineer that understands your board



Baker Studio today: Proto found the thin trace, fixed it and planned the mill. You press Accept.

01

READS

schematics and layouts, via the Flux.ai API

02

MAKES

the Baker job: mill, drill, mask, stencil, jig

03

DIAGNOSES

through Mixer, in plain words

04

HAS A CHARACTER

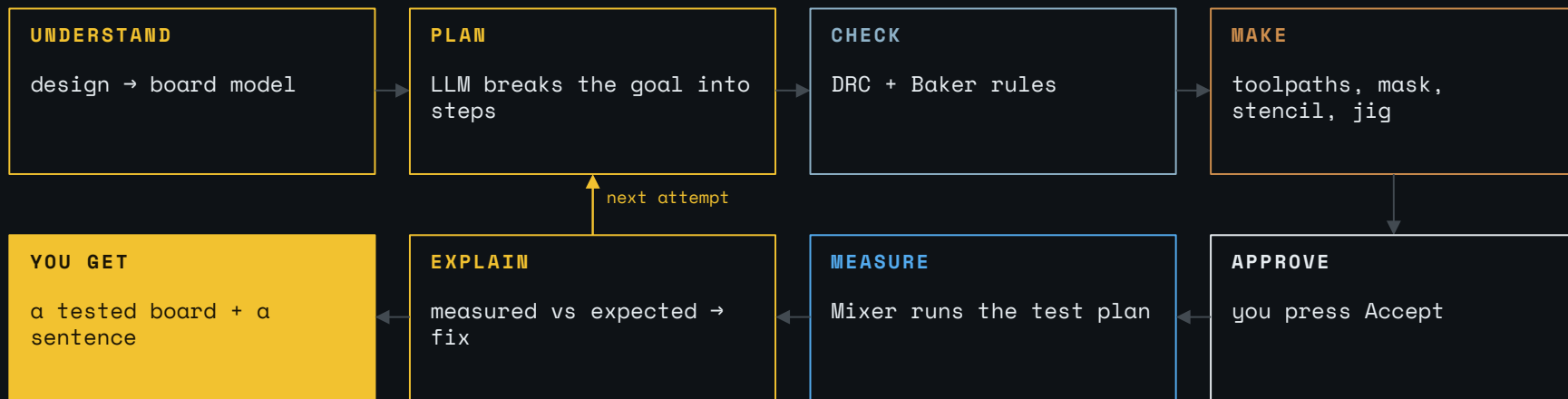
a platypus engineer, built on Gemini

NEXT

Probe matrix: probes under the board reach every net.

Simple enough that a **high-school student** can print a board.

A loop, not a chatbot



MEMORY every run logs design → settings → measurements → outcome. No files-only AI has this data: it's the moat.

The model proposes · tested engines compute · you approve · the machine executes.

From “import” to “PASS” in one sitting

baker studio — sensor-hat-rev-c.brd

10:02	you	import sensor-hat-rev-c from Flux
10:03	proto	one trace under the MCU necks to 5.8 mil. Widened to 7 mil. [Accept]
10:04	baker	mill · drill 42 holes · mask · stencil · jig prints in parallel
11:40	you	drop parts into the jig
11:52	baker	low-temperature reflow
12:05	mixer	power up ✓ 3V3 rail ✓ I ² C sensor: no reply ✗
12:08	proto	“SDA has no pull-up: R4 is lifted on one pad. Reheat R4.”
12:15	mixer	retest: sensor replies ✓ PASS

Illustrative run on the Baker Studio board. Times are engineering targets; the proof of concept measures them.

Fifteen steps on one shuttle, in one box



≈2 h

per board

15–20 min

of it hands-on

3–5

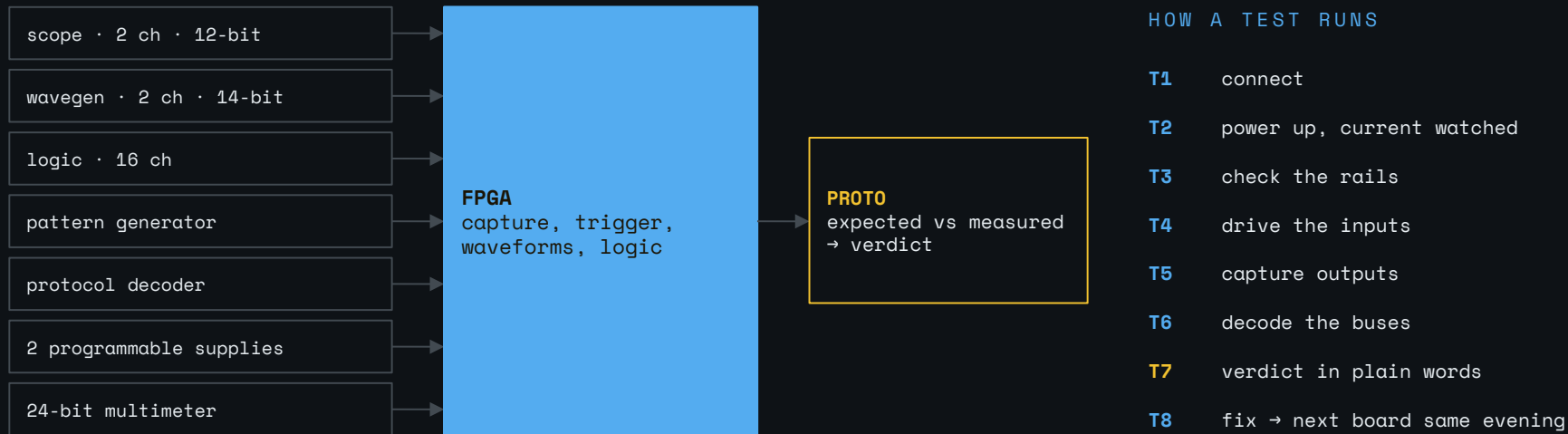
boards per shift

₹1.45L

production cost

Times are engineering targets. Prototype ₹3,73,418 → production ₹1,44,842 (China parts, Delhi build, 100 units).

Eight instruments, one cable, one brain



₹13,740

production cost

8 in 1

scope to 24-bit meter, AD3 class

Our tester

on the China line too

Prototype ₹74,847 keeps 12/14-bit converters; production drops to 8-bit, one USB cable, no screen. Status: designed, first build in the proof of concept.

Others solve one step. Nobody closes the whole loop.

PRODUCT	PRICE · ₹	MAKE	ASSEMBLE	TEST	GUIDE
Voltera V-One	~₹3.1 lakh	●	●	○	○
LPKF ProtoMat	8.6–24.5 lakh	●	○	○	○
Makera Carvera	~₹5.1 lakh	●	○	○	○
Bantam Tools · discontinued	~₹3.1 lakh	●	○	○	○
Digilent Analog Discovery 3	~₹33,000	○	○	●	○
JLCPCB, PCBWay	per order, weeks	●	●	○	○
Flux (AI design)	1.3–4.3k / month	○	○	○	●
protopus	3.34 lakh	●	●	●	●

List prices at ₹88 per US\$, before import duty · [voltera.io](#) · [Design World](#) · [Makera](#) · [ToolGuyd](#) · [Digilent](#) · [flux.ai](#)

We earn on every machine. The customer earns it back in months.

PER UNIT · ₹

BAKER

MIXER

Total cost (all-in)	2,00,000	22,500
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Selling price	2,99,000	34,999
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Margin	~33%	~36%
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₹9,999 / year · one Proto plan per workspace

+ consumables, reordered as labs build

FOR THE CUSTOMER

₹334

per board on Baker

₹6,721

same board from JLCPCB, landed

53 boards

to pay back Baker + Mixer: about 6.6 months for a startup
iterating twice a week

Start in the maker studios, then follow the people who watch them

₹6.4L cr global PCB market	₹3.1L cr global test and measurement	₹55K cr India PCB, ~15% a year	₹8.3K cr India test and measurement
33M+ active Arduino users	2.3M+ Hackster hardware community	10,000 Atal Tinkering Labs, +50,000 planned	~2,500 Fab Labs in 125 countries

NOW

Maker studios

Free units for featured videos. The Bambu Lab playbook.

NEXT

Hobbyists and startups

Everyone who watches and loses weeks per board.

ALONGSIDE

Prototyping labs

Tinkering, Fab and college labs that already buy tools.

Told honestly, one milestone at a time

Baker

prototype running on the bench

Software

Baker Studio + Proto; website live

Mixer

designed, not yet built



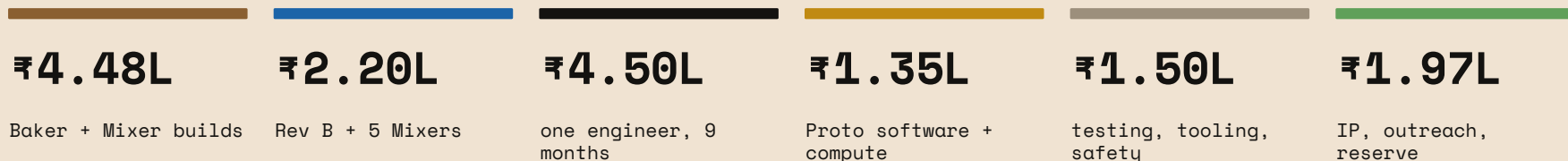
SIGNAL

Samuel Beek, Founder & CEO of Schematik, is interested in funding us at proof of concept.

Built in the university lab, tested with students and makers. A demo video every month.

₹16,00,000

nine months to a proof of concept, and time to raise.



Enough to survive one failed iteration and raise from strength. Builds line by line from the cost report.

After proof of concept: raise, build in China, run lean from Delhi

THE ROUND

~₹1.5 Cr seed

China pilots ₹70L · set-up + IP ₹25L · team of 5 ₹25L · Delhi centre ₹20L · buffer ₹10L

THE CHINA PLAN

Design in Delhi, build in Shenzhen

Mixer: turnkey from batch one. Baker: modules first; turnkey once ≤ \$1,305 ex-works

OPERATIONS

Delhi finalises, doesn't build

Scan, test, activate Proto, ship. Mixer is our production tester: our rig, our data

MONTH 0-2

design freeze, 12 RFQs

MONTH 3-5

EVT + DVT pilots

MONTH 6-9

PVT: 100 Mixers, 20 Bakers

MONTH 10-12

first container



Thank you.

Come and try the loop.

protopus-kappa.vercel.app

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MANAGEMENT TEAM & CORE SKILLS

Who is building this

Soham

Founder

Third year ECE, Faculty of Technology, University of Delhi

Industrial design

Mechanical architecture

Embedded electronics

Machine software

HIRING NEXT · a software engineer who also writes firmware.

